A

AthletixAI

AI-powered sports talent assessment — using nothing but a phone camera.

Archita Tripathi · **Manthan Dhanraj**

athletixai.onrender.com · live in production

35
SPORTS

33
KEYPOINTS

5
METRICS

THE PROBLEM

India doesn't lack talent. It lacks measurement.

No coach

A rural athlete has no trained eye to tell them what to fix.

No baseline

Without numbers, nobody knows if they're improving or stuck.

No visibility

Selectors can only pick athletes they've physically seen.

Why this exists

A professional biomechanics lab costs lakhs and exists in only a handful of cities — so most athletes are never measured at all. AthletixAI was built so that fewer athletes lose their chance for lack of resources. It's free for athletes, always.

THE SOLUTION

A performance lab that fits in a pocket.

Record a clip on any phone. AI analyses it on the device and returns a report in seconds — scored across five metrics, tuned to your sport.

Speed	Peak burst + cadence
Agility	Symmetry + direction change
Strength	Joint depth + range
Stamina	Sustained output
Technique	Control + form quality

What makes it different

- Real computer vision — not a chatbot wrapper
- Video never leaves the phone
- Scoring tuned per sport, not one formula for all
- Refuses to score a clip it can't actually read
- Free for athletes, always

HOW IT WORKS

From a phone clip to a scored report

1 Record

Athlete films a drill on any phone, or uses live camera mode.

2 Detect

MediaPipe extracts 33 body keypoints from every sampled frame.

3 Measure

Joint angles, velocity, symmetry, cadence and range of motion.

4 Score

A sport-category formula turns those into five metric scores.

5 Report

Insights, weak-area drills, injury zones and a progress trend.

6 Share

The athlete's coach sees the report instantly on their dashboard.

THE AI

Real machine learning, real measurement

Joint angle from three keypoints

$$\cos(\theta) = (v1 \cdot v2) / (|v1| \times |v2|)$$

Knee angle = hip → knee → ankle. A straight leg ≈ 180°, a deep squat ≈ 70°. The same routine gives the bowling-arm angle.

No random numbers

The scoring path is fully deterministic — the same clip always produces the same score, a property we test for directly.

1

Pose estimation

MediaPipe returns 33 landmarks per frame — each with an x, y and visibility.

2

Feature extraction

Angles, velocity, symmetry, cadence & range accumulate across the clip.

3

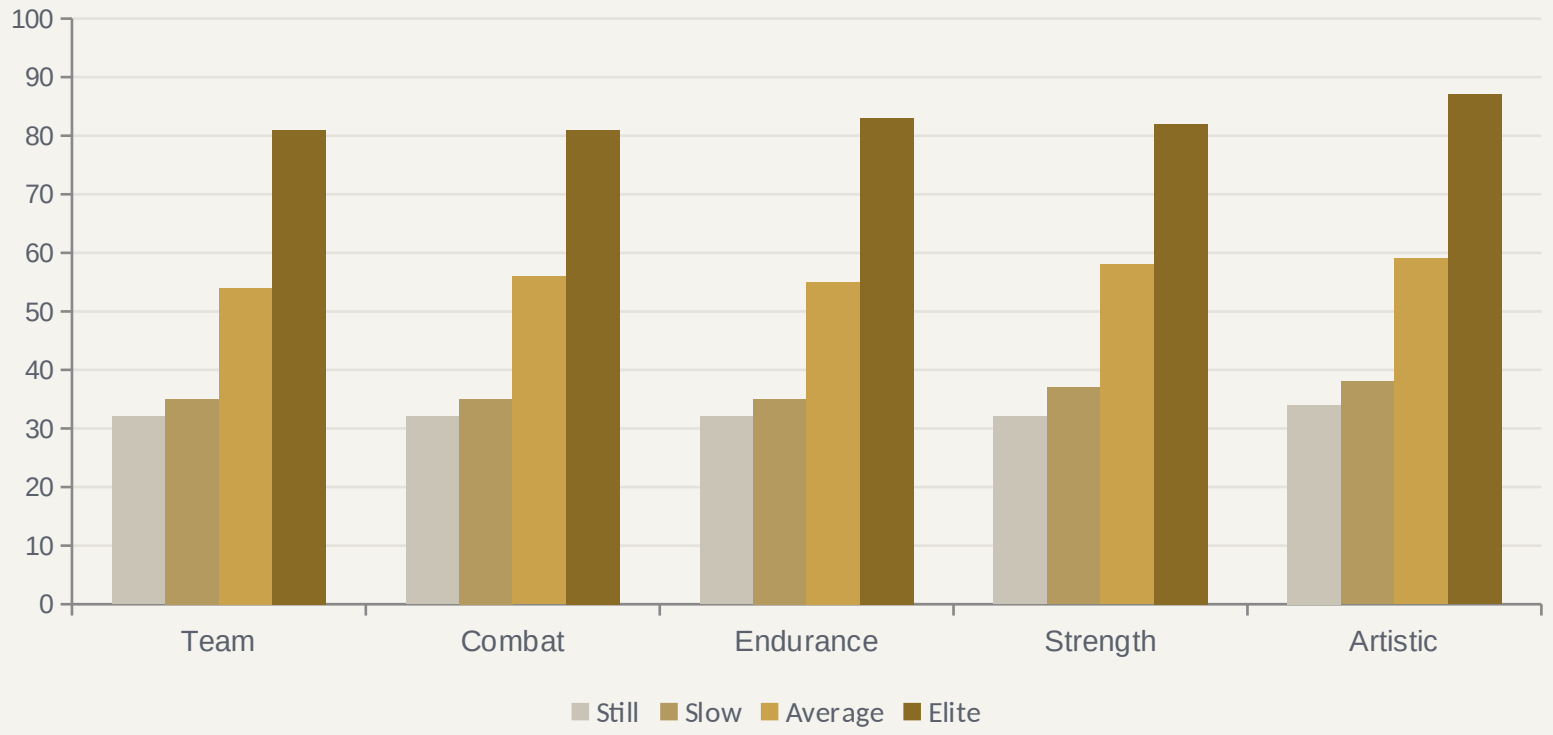
Category scoring

Six profiles map features to 5 metrics; an effort gate blocks low-effort highs.

VALIDATION

Effort has to move the score

Four synthetic athlete profiles were fed straight into the engine. A correct engine must rank them: still < slow < average < elite.



Result

All 8 categories ranked correctly. Precision is intentionally flatter — stillness is skill there.

A bug this caught

A slow-jogging clip once scored 99/100 — the scorer was using on-screen defaults, not the video. Fixed: it now scores 42.

HONESTY BY DESIGN

The system refuses to guess

A tool that invents a number when it can't see the athlete is worse than no tool at all. So the engine returns an honest error instead of a score when:

Fewer than 12 frames with a detected person	*“No clear person could be detected.”*
Person visible in under 40% of frames	*“Keep the athlete fully in frame.”*
Not enough measured movement data	*“Upload a longer, clearer clip.”*
Low tracking confidence (blurry / dark)	*Score is capped, never inflated.*

COVERAGE

35 sports — every one genuinely analysable

Sports a phone camera can't honestly read were removed rather than shipped unreliable — water, winter, equestrian and mind sports.

Team	9	Football · Basketball · Volleyball · Rugby · Hockey · Handball · Lacrosse · Kabaddi
Combat	7	Boxing · Wrestling · MMA · Judo · Karate · Taekwondo · Fencing
Bat & Ball	3	Cricket · Baseball · Softball
Racquet	3	Tennis · Badminton · Table Tennis
Endurance	2	Athletics · Cycling
Precision	5	Archery · Shooting · Golf · Snooker · Billiards
Artistic	5	Gymnastics · Parkour · Climbing · Skateboarding · Breakdancing
Strength	1	Weightlifting

Removing 13 sports was an integrity decision, not a gap.

ENGINEERING

How it's built, and at what scale

Browser

`index.html`

Single-file UI · 159 JS functions · MediaPipe runs here

→

Flask API

`app.py`

31 REST endpoints · auth · CSRF · rate limiting

→

SQLite

`athletixai.db`

12 tables · cascading deletes · indexed reports

The browser never touches the database — it only calls the API. That's what keeps the data layer safe.

5	31	12	159	35	33
Languages	Endpoints	Tables	JS functions	Sports	Keypoints

Vanilla JS · Flask · SQLite · MediaPipe · Chart.js · bcrypt · PWA · 3-language UI (English / हिन्दी / Hinglish)

Free for athletes. Always.

Talent should not depend on where you were born or who happened to be watching.

Live	**athletixai.onrender.com**
Code	**github.com/manthandhanraj/AthletixAI**
Built by	**Archita Tripathi · Manthan Dhanraj**

Built with AI assistance.

The architecture, product decisions, debugging and deployment are ours — and we can walk through any line of it.

Thank you.